Seatwork 3.1	
Creating a Basic Face Filter	
Course Code: CPE 411	**Program:** Computer Engineering
Course Title: Interactive Extended Reality	**Date Performed:** 19/02/2026
Section: 22S2	**Date Submitted:** 19/02/2026
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1. Discussion

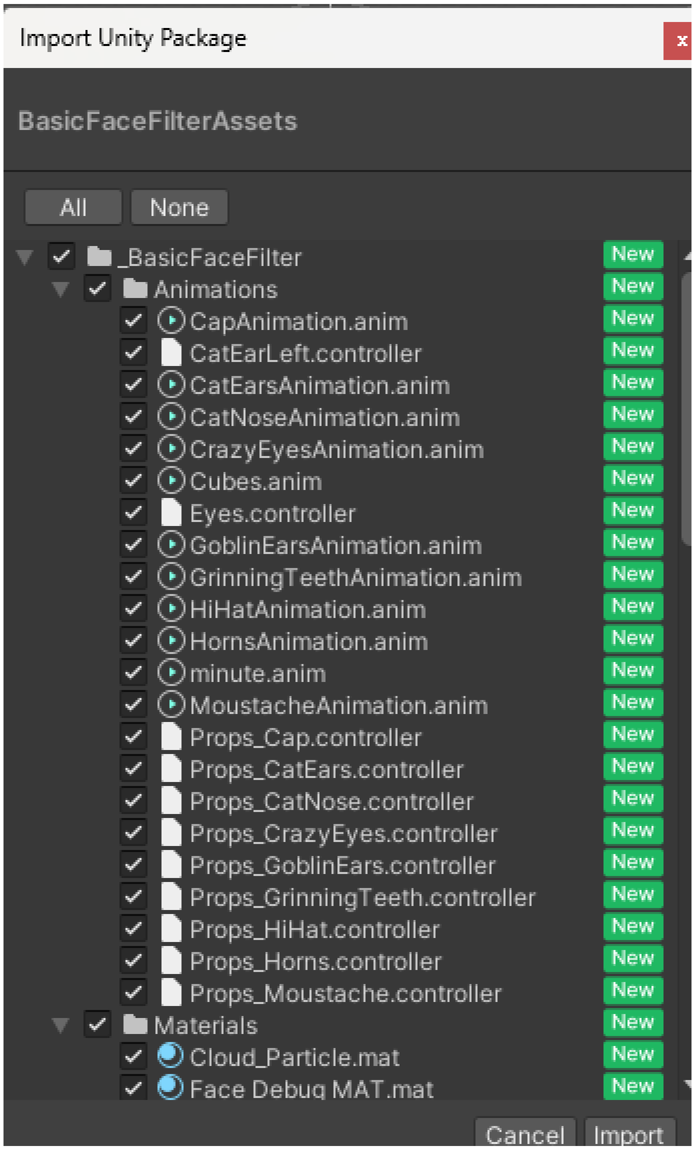
This activity aims to hone the skills of the student in using Unity's ability to build AR technology by creating a basic face filer similar to Tiktok or Instagram filters

2. Materials and Equipment

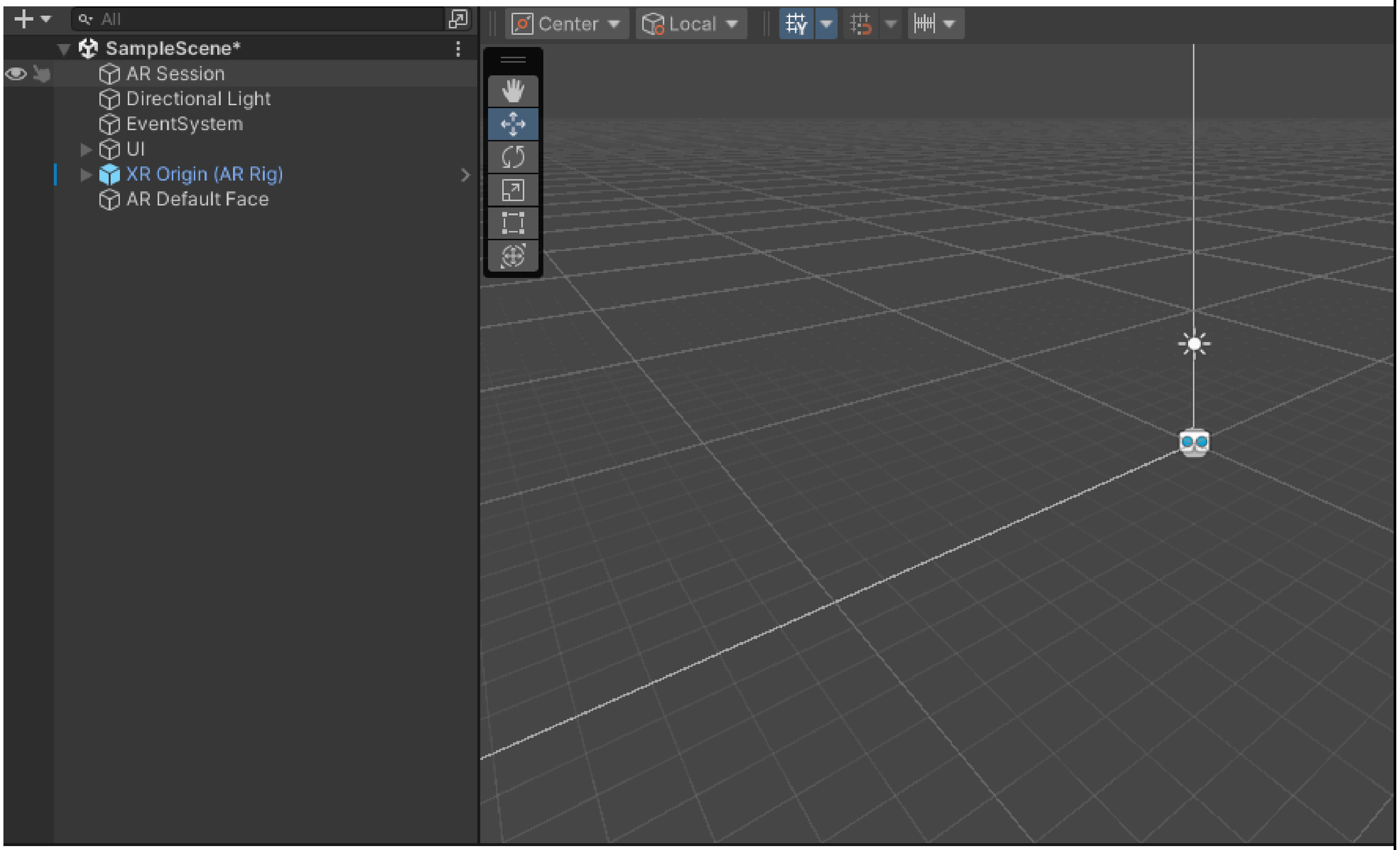
Unity Hub and Unity Editor
Learn Unity Website (Unity Essentials Pathway - Learn Game Development for Beginners | Unity Learn)
Computer with Network Connection

3. Procedure

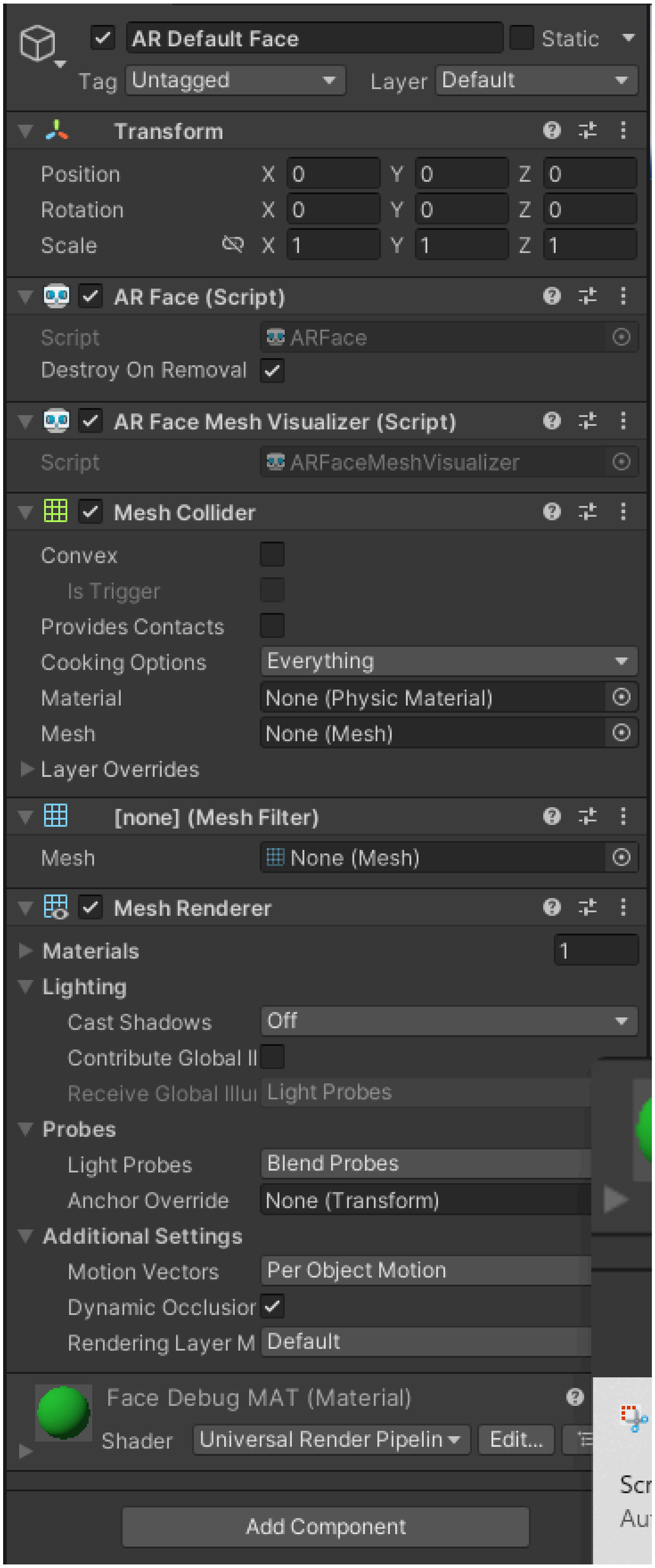
Open the AR Project template, named it "BasicFaceFilter"



Import the basic face filter assets, containing controllers, animations, and the assets / visuals, as it seems.

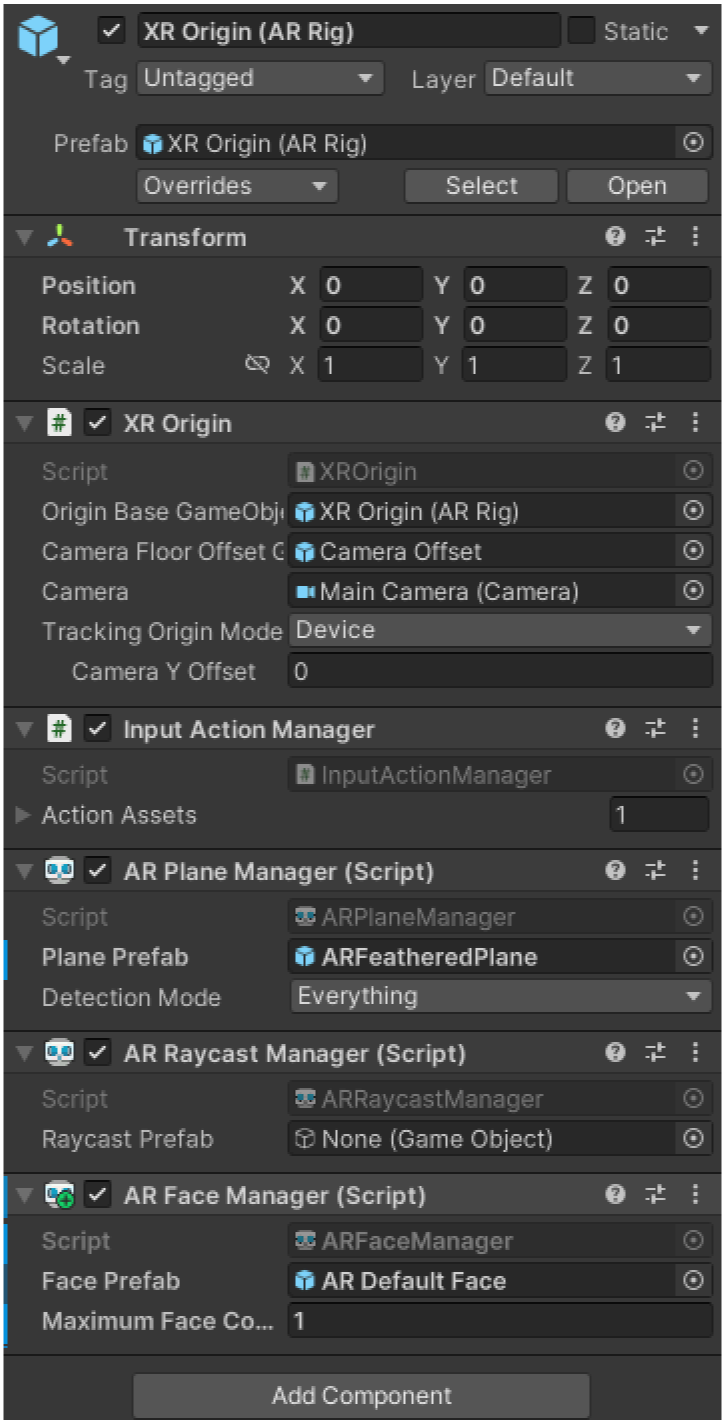


Created an AR Default Face object as instructed. This AR Default face object serves as the customizable object I can tinker with in order to make my own personalized filters

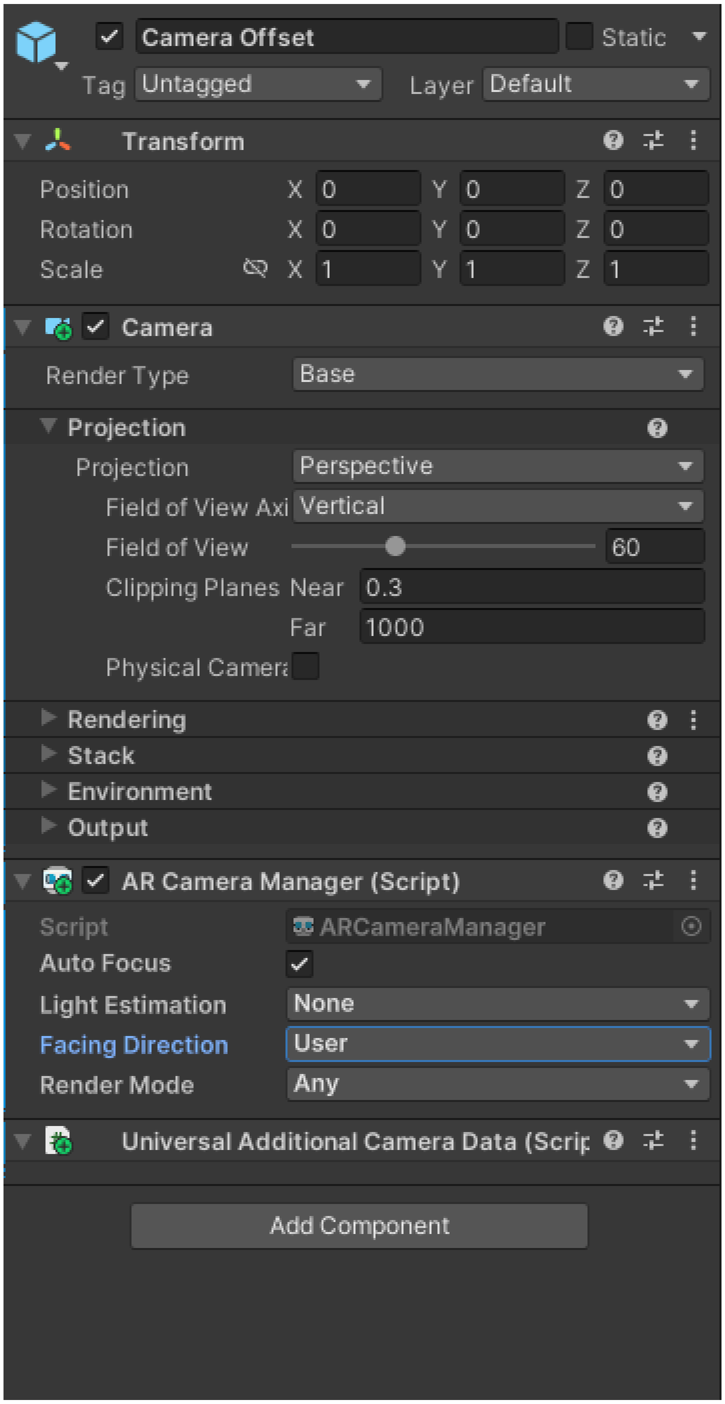


Set the debug material as the face’s material for easier testing

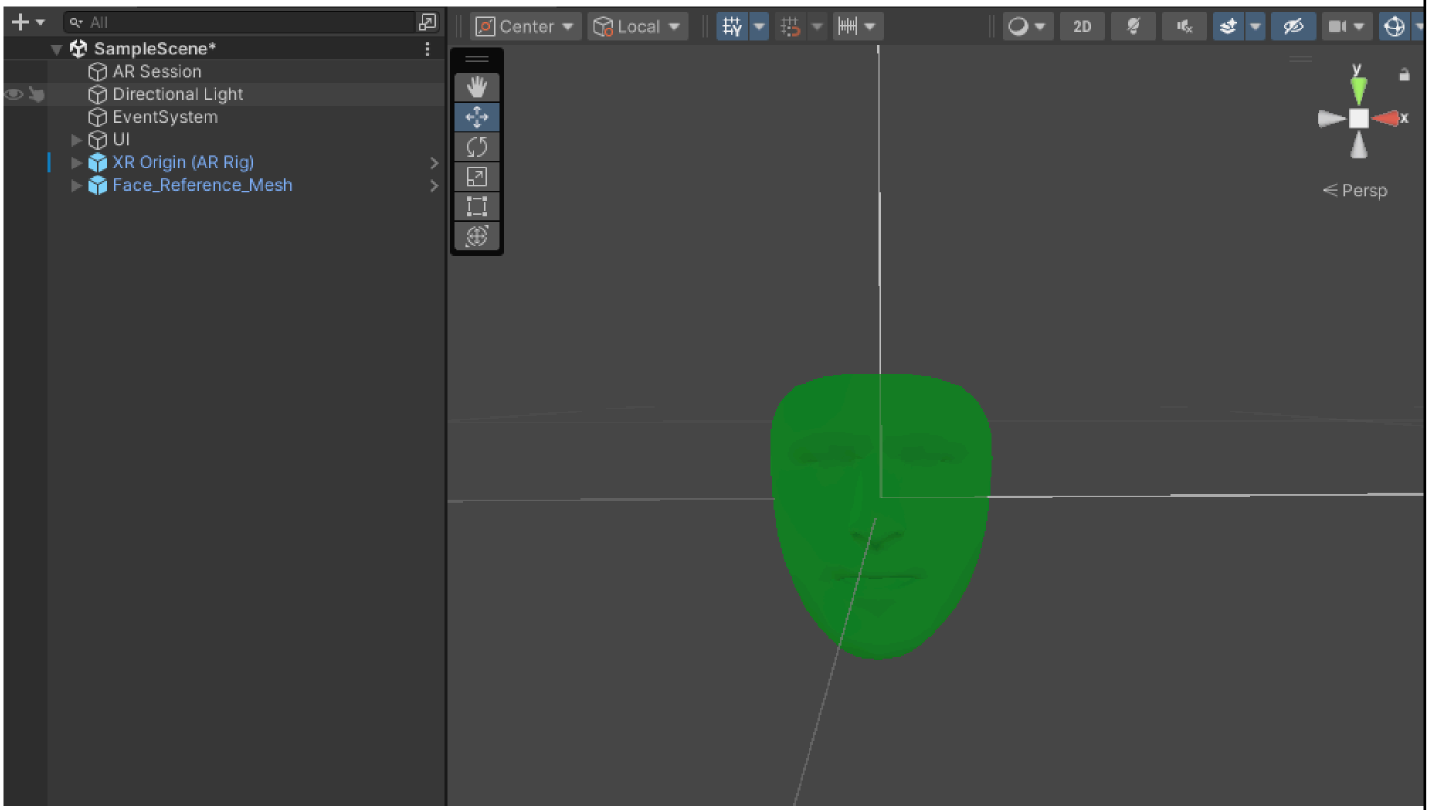
Added the default face as a Prefab material



Add a face manager script to the AR Rig object, and set the default face object prefab we created previously as the face that the script would manage

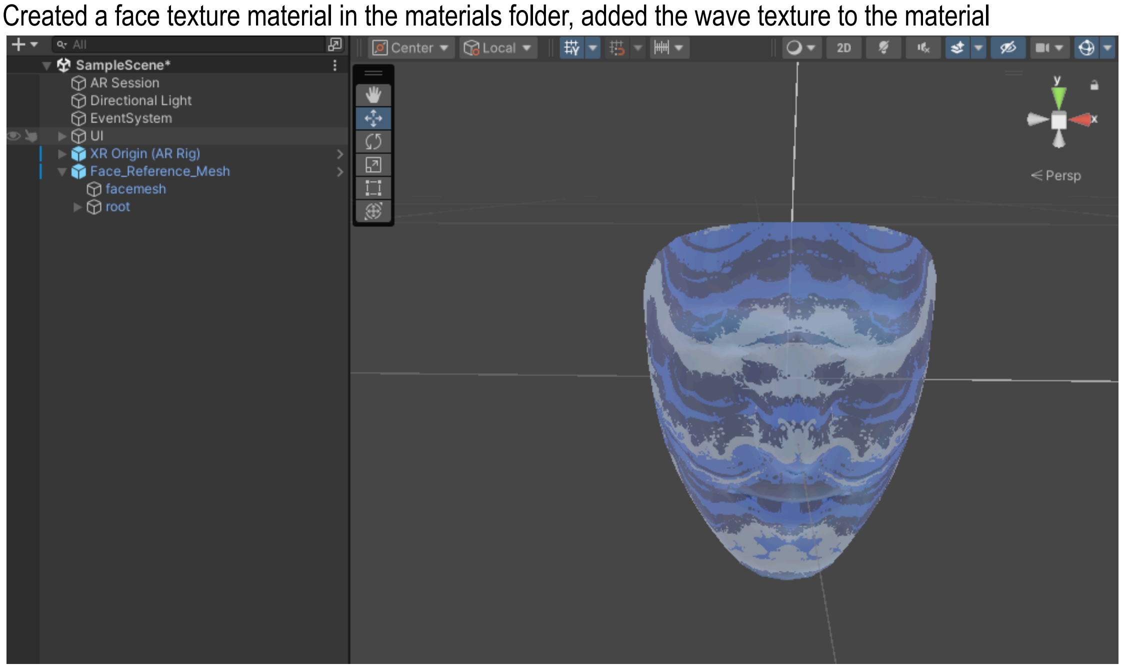


Add an AR Camera manager script to the camera offset child of the AR Rig, then set the facing direction to the user

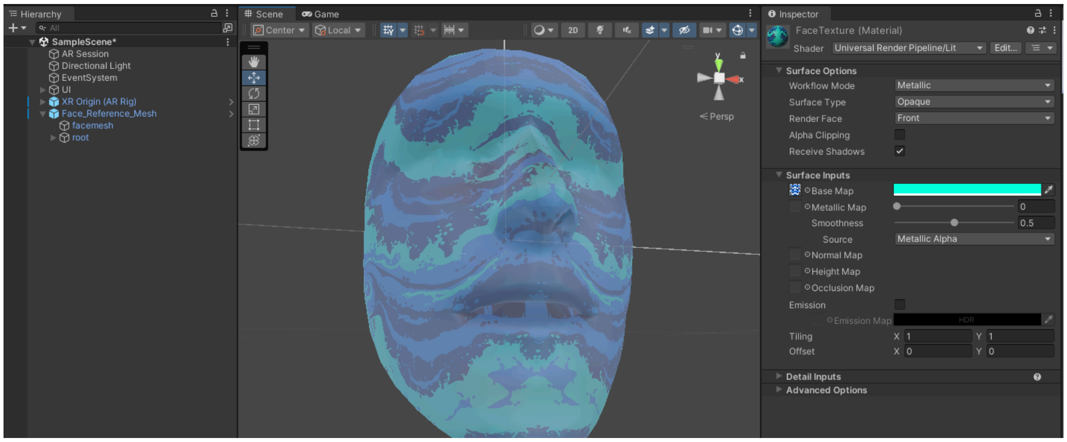


Dragged the FaceReferenceMesh to the hierarchy

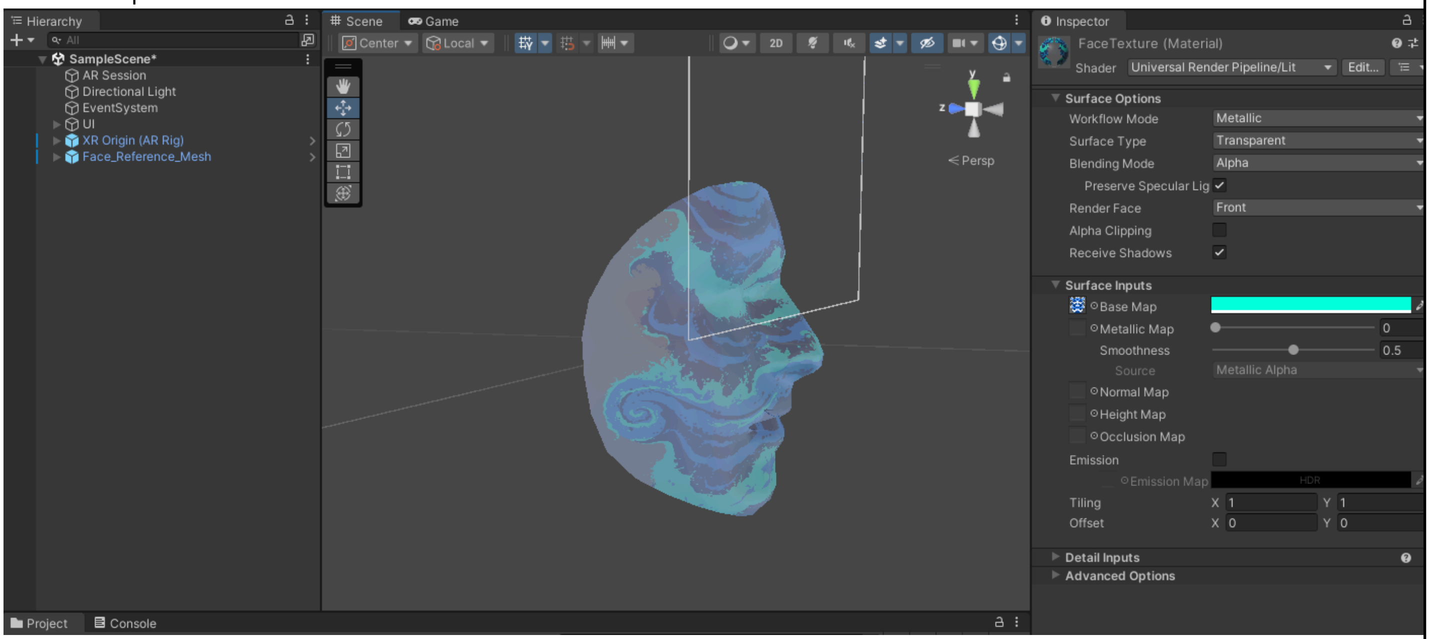
Created a face texture material in the materials folder, added the wave texture to the material



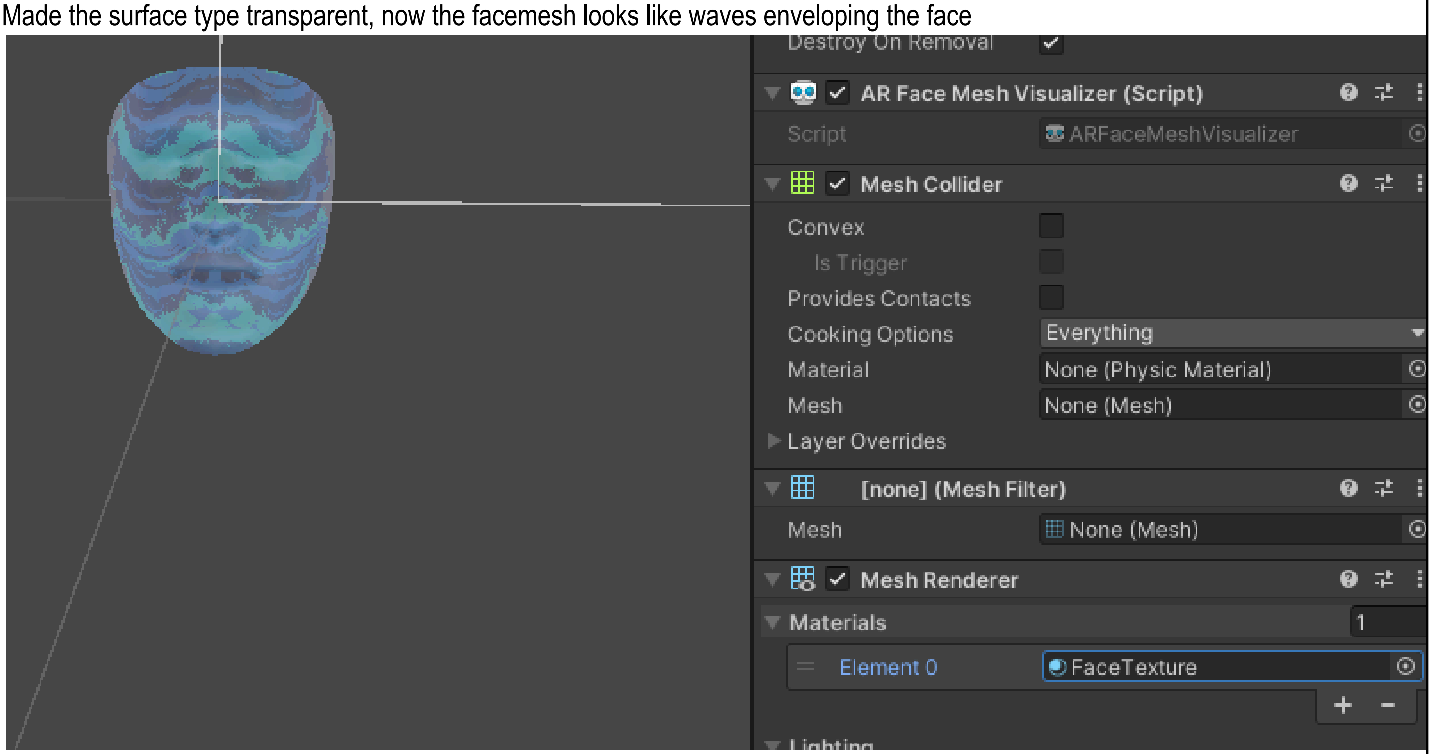
Dragged the material to the facemesh child object, applying the texture to the facemesh



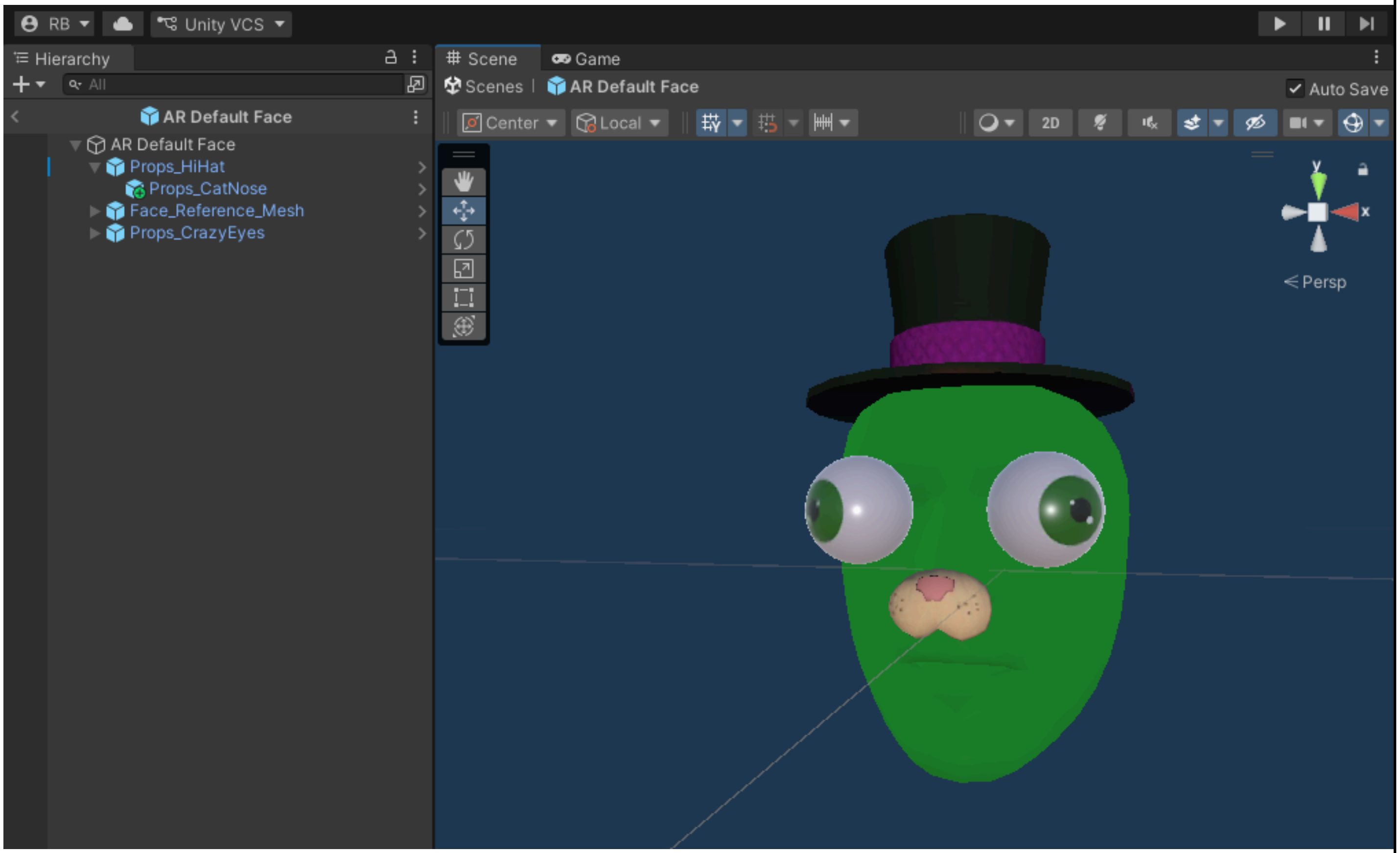
Added a turquoise tint to the mesh



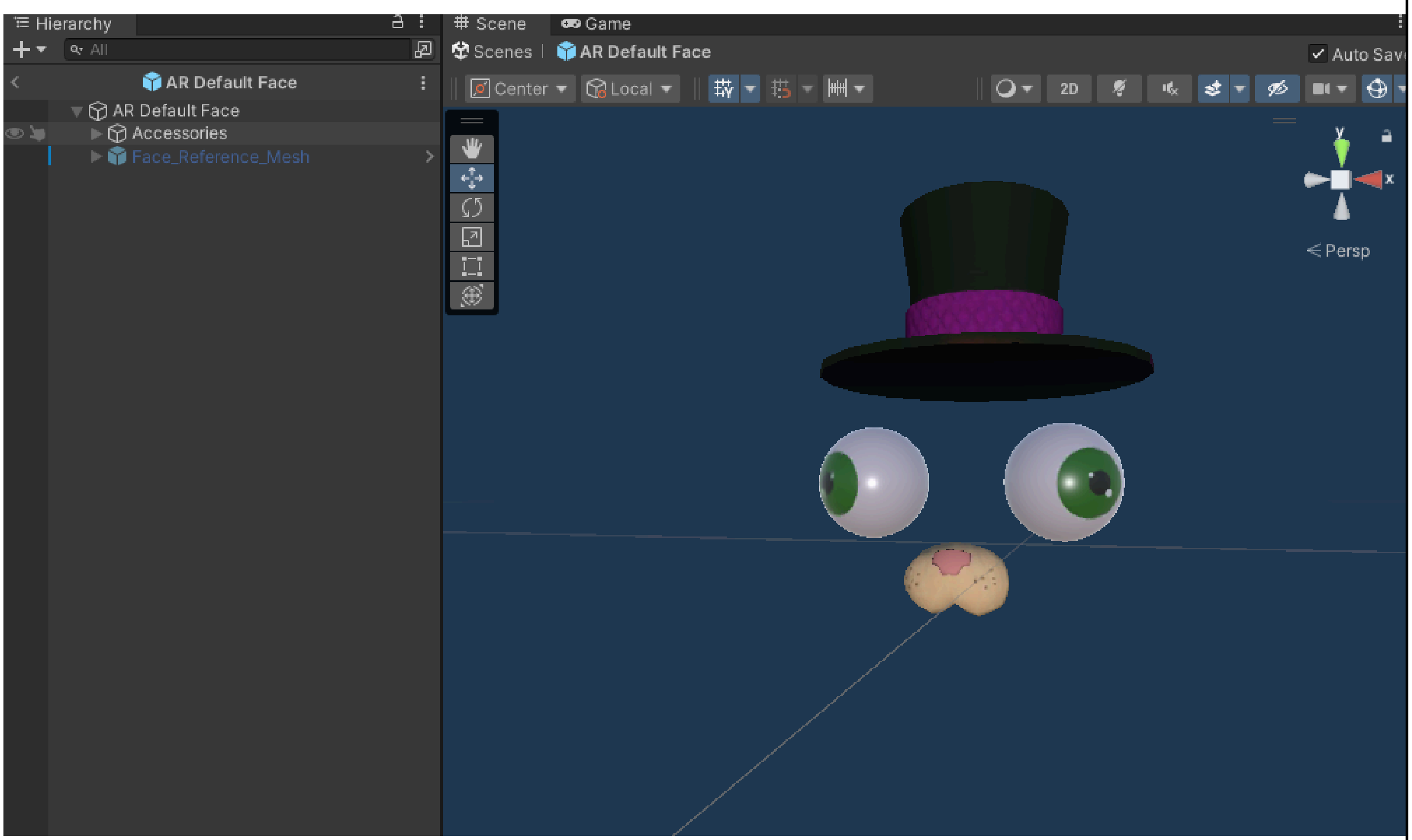
Made the surface type transparent, now the facemesh looks like waves enveloping the face



Alter the AR Default Face prefab so it uses the newly made facetexture as the material

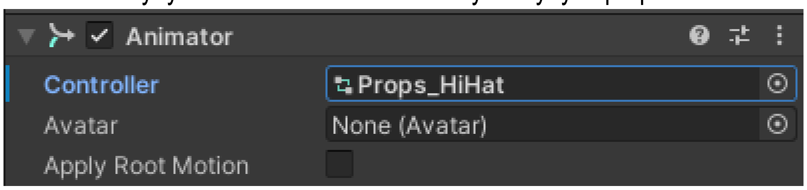


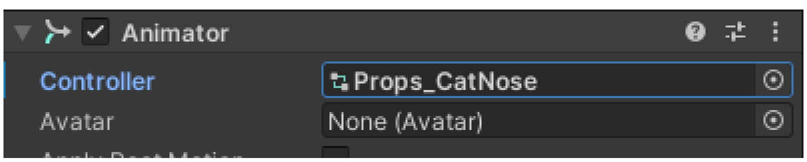
Modifying the default face, I first added a reference mesh for visualization, then added accessory prefabs to the face.



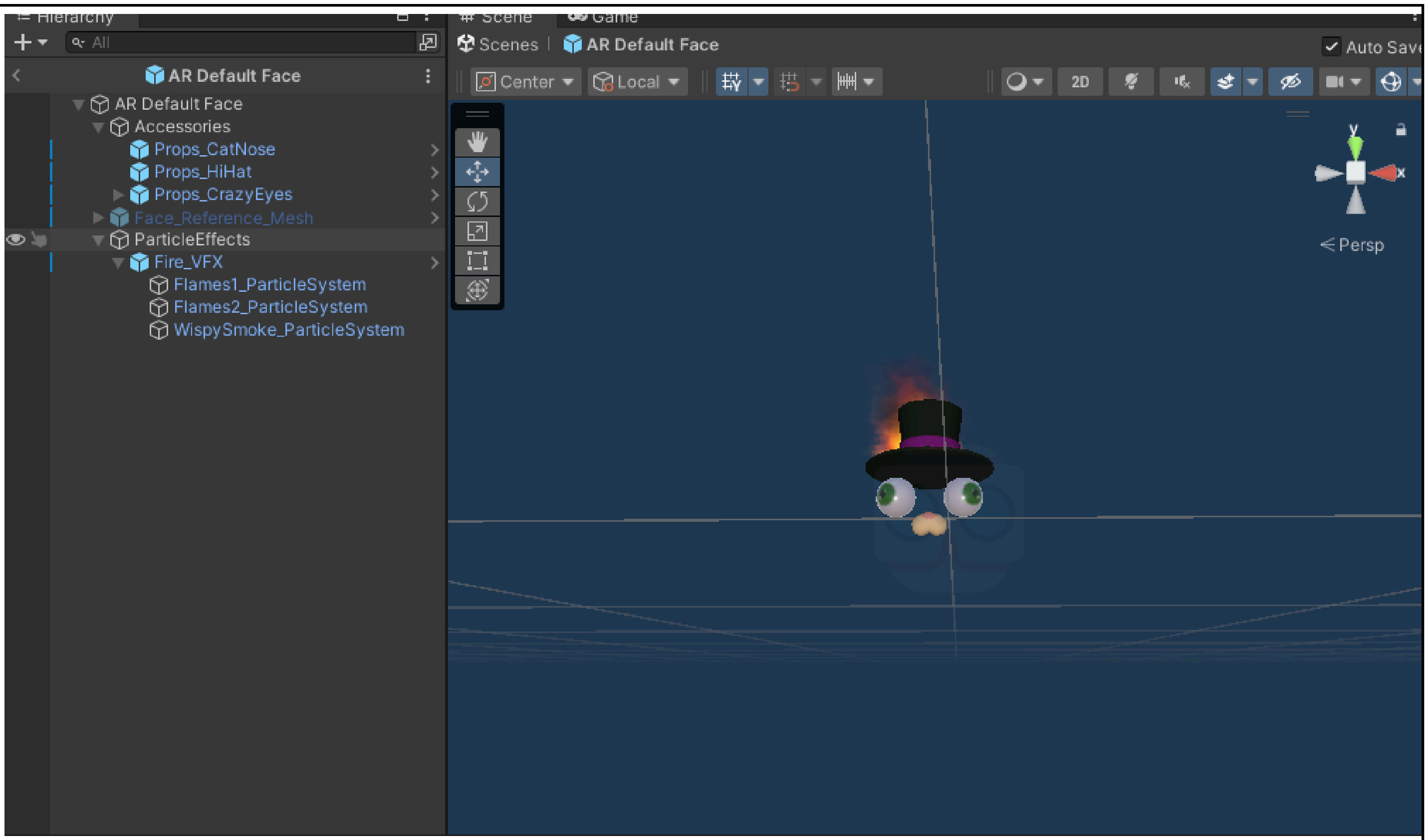
I disabled the reference mesh, and this is the final result of my customization

Added a crazyeyes animator controller for my crazyeyes prop

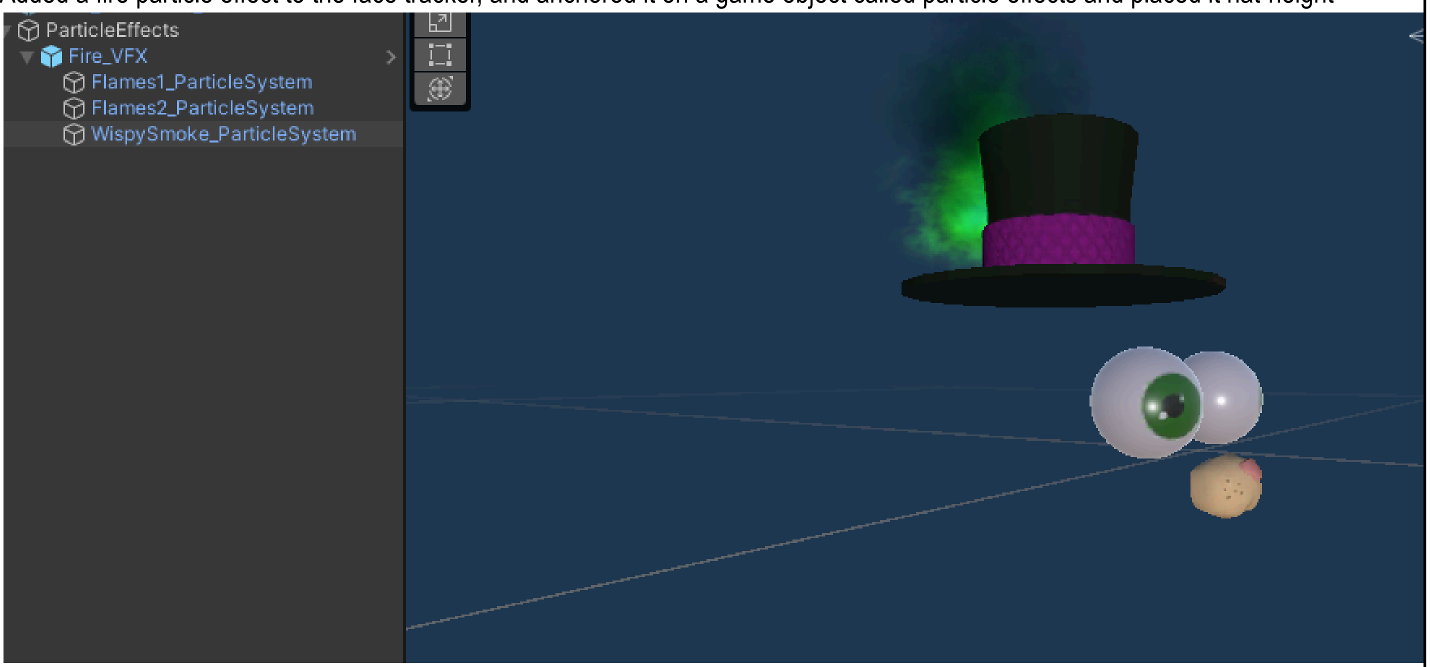




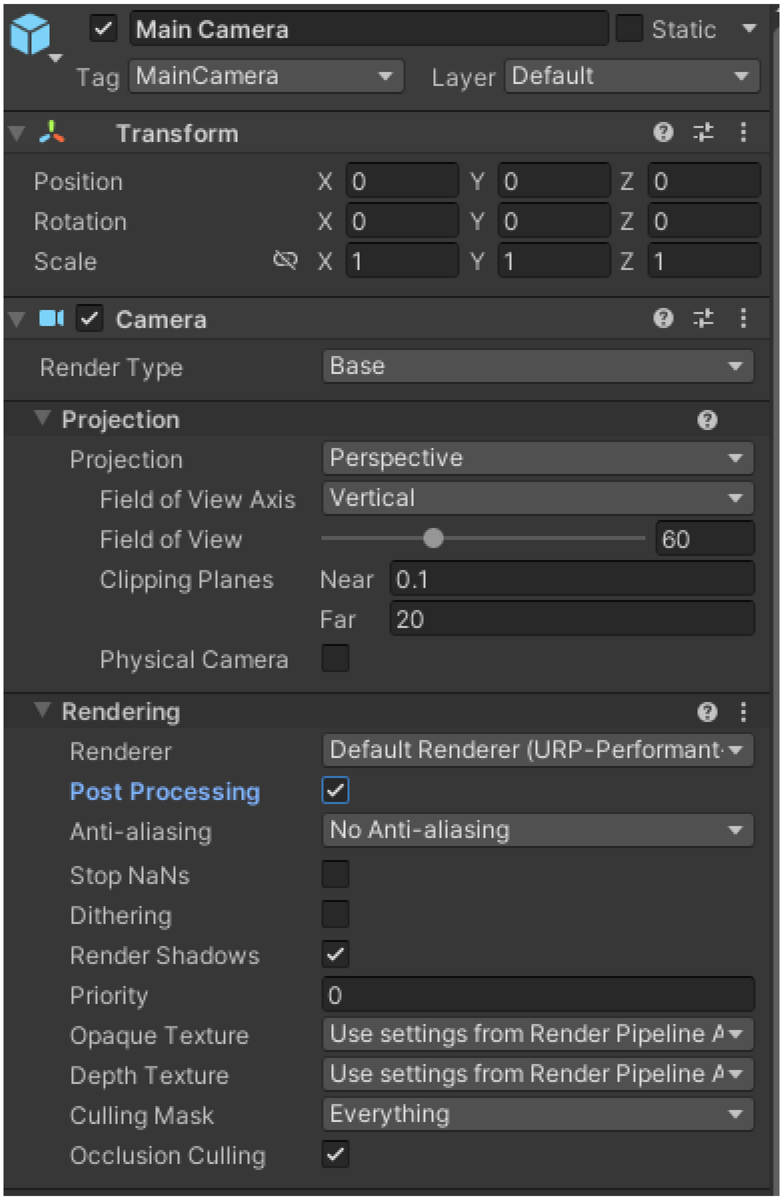
I also added animations to the hi_hat and catnose; the other accessories



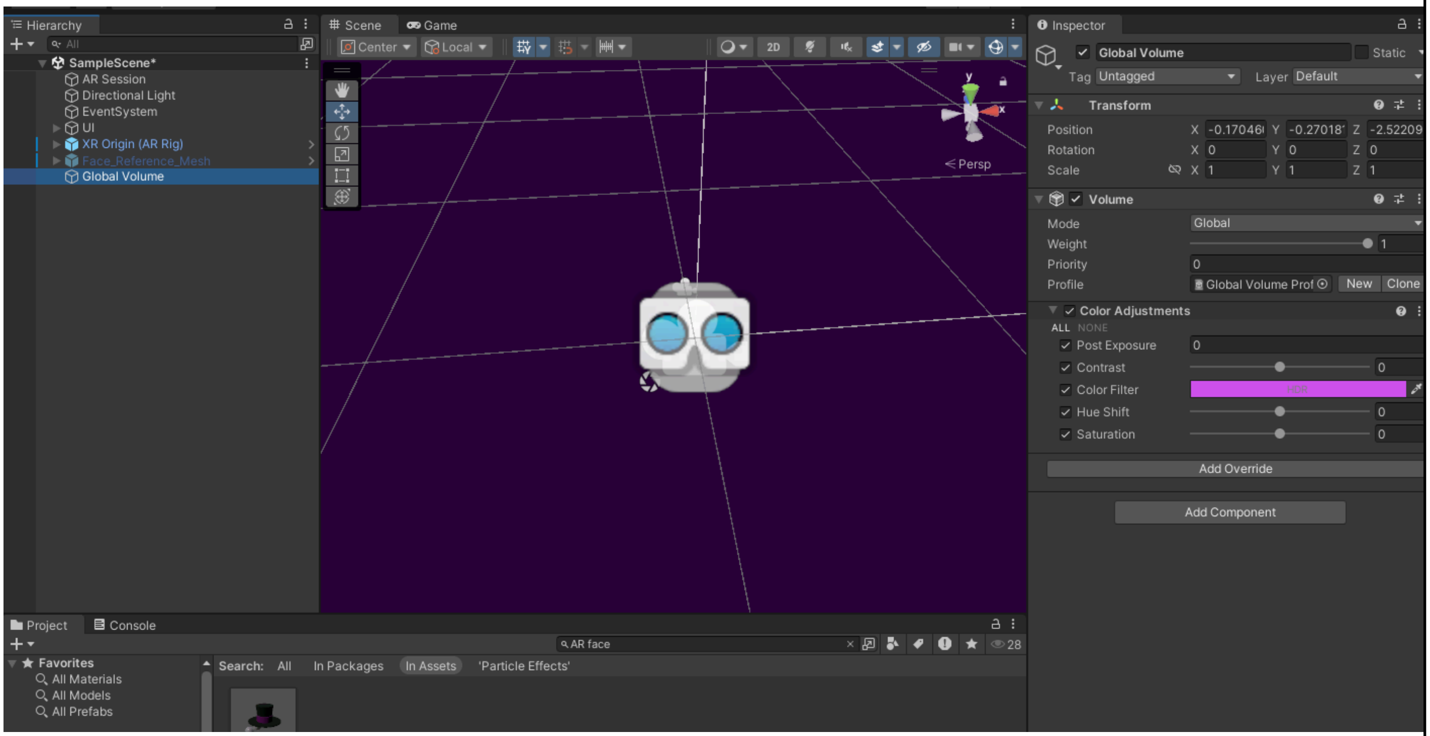
Added a fire particle effect to the face tracker, and anchored it on a game object called particle effects and placed it hat-height



Customizing the particle system, I made the fire green (simulation space set to world as instructed)



Ticked the post-processing option in the Main Camera of the AR Rig's Camera Offset



Added a post-processing volume override, changed the "filter" color to a purple/pinkish color, the direct contrast of green (from my green flame animation)

I asked for my classmate's help for building as I do not own an Android device capable of testing AR applications

4. Conclusion

After this activity, I learned a lot about face filters and what I can build using Unity's technologies. I applied materials, added accessories, effects, color gradings, and animations. I did not struggle nor find any part of the activity confusing and/or beyond my current skill level, so I easily caught up with the tutorial and was able to build my own simple face filter. In the end, I am sure that the skills I learned within this activity will serve a good purpose, and I'll be able to build my own face filters in the future..

5. Assessment Rubric

Lab Activity Rubric

Criteria	Ratings						Pts
SO 7 PI 1 **Student Outcome 7.1** Acquire and apply new knowledge from outside sources. threshold: 4.8 pts	6 pts Excellent | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently	4 pts Satisfactory | Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory | Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor | Relies on classroom instruction only	1 pts Very Poor | No initiative or interest in acquiring new knowledge	6 pts
SO 7 PI 2 **Student Outcome 7.2** Learn independently threshold: 4.8 pts	6 pts Excellent | Completes an assigned task independently and practices continuous improvement	5 pts Good | Completes an assigned task without supervision or guidance	4 pts Satisfactory | Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory | Requires detailed or step-by-step instructions to complete a task	2 pts Poor | Shows little interest to complete a task independently	1 pts Very Poor | No interest to complete a task independently	6 pts
SO 7 PI 3 **Student Outcome 7.3** Critical thinking in the broadest context of technological change threshold: 4.8 pts	6 pts Excellent | Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good | Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory | Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory | Apply the gathered information to formulate the problem	2 pts Poor | Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor | Gather information from a variety of sources	6 pts
SO 7 PI 4 **Student Outcome 7.4** Creativity and adaptability to new and emerging technologies threshold: 4.8 pts	6 pts Excellent | Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good | Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory | Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory | Shows some creative ways to solve the problem	2 pts Poor | Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor | Ideas are copied or restated from the sources consulted	6 pts
							Total Points: 24